BHSMS System Guide

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A complete guide to the Barangay Health Status Monitoring System — every page, role, clinical threshold, and workflow for the Municipality of Bucay, Abra.

System Overview

BHSMS is a Next.js web application with a Supabase (PostgreSQL) backend, built for offline-first operation in rural Philippine barangays where internet connectivity is intermittent. Patient data is cached in the browser's localStorage and synced to the cloud when a connection is available.

The system covers three barangays: **Palaquio**, **Tabiog**, and **Babangcag** in the Municipality of Bucay, Abra.

Every registered resident receives a **Unique Patient Identifier (UPI)** in the format `BHS-[PAL|TAB|BAB]-[YEAR]-[SEQUENCE]`. The system uses **Soundex deduplication** to prevent duplicate registrations of Filipino names (e.g., "Dela Cruz" vs "De la Cruz" are recognized as the same person).

All data access is logged in an audit trail for compliance with Republic Act No. 10173 (Data Privacy Act of 2012). Row-Level Security (RLS) on all database tables ensures users can only access data their role permits.

User Roles & Access

The system has 5 roles, each with different module access controlled by modular RBAC (Role-Based Access Control). Admins can override defaults to grant or restrict specific modules per user.

BHW (Barangay Health Worker)

- **Example user:** Elena Bagani — Barangay Palaquio BHS
- **Default modules:** Resident Registry, New Patient Intake, Vitals & Triage, Child EPI Immunization
- **Primary job:** Community-level registration, basic vital signs capture, immunization outreach, and defaulter follow-up in the purok

Nurse / Midwife

- **Example user:** Corazon Valdez, RM — Barangay Tabiog BHS/RHU
- **Default modules:** Everything the BHW has, plus Maternal Care (ANC) and Hospital Referrals
- **Primary job:** Clinical vitals with triage evaluation, maternal prenatal tracking, referral slip generation
- **Special permission:** Can be granted elevated access to the Medical Patient Chart by an admin

Doctor (Municipal Health Officer)

- **Example user:** Dr. Arthur Alcantara — Bucay Municipal Health Office
- **Default modules:** Resident Registry, Vitals & Triage, Medical Patient Chart, Maternal Care, Child EPI, Hospital Referrals, Epidemiology Analytics
- **Primary job:** Clinical consultations, ICD-10 diagnoses, prescription management, surveillance analytics review

LGU Viewer (Local Government Unit)

- **Example user:** Hon. Ramon Balbin (Barangay Captain) — Tabiog Council Hall
- **Default modules:** Epidemiology Analytics (view-only), Workstation Operations & Backup
- **Primary job:** View aggregated population health reports and system status. Cannot see individual patient records or clinical data.

Admin (System Administrator)

- **Example user:** LGU System Administrator — Bucay IT & Health Ops Center
- **Default modules:** All 10 modules, including User Management
- **Primary job:** Provision user accounts, configure per-user module permissions, manage cloud sync, run system diagnostics

Pages & Modules

The system is organized into 10 modules across three categories. Access is enforced by the ModuleGuard component and Supabase Row-Level Security.

Frontline & Community Intake

Page	Route	Roles	What It Does
Resident Registry	`/residents`	BHW, Nurse, Doctor, Admin	Searchable demographic database with filters and Medical Chart modal access
New Patient Intake	`/residents/register`	BHW, Nurse, Admin	Register new residents with demographics, household, PhilHealth ID. Live Soundex warns of duplicates
Vitals & Triage	`/vitals`	BHW, Nurse, Doctor, Admin	Record BP, HR, RR, temp, weight, height. Auto-calculates BMI and flags abnormal readings

Clinical & Specialized Care

Page	Route	Roles	What It Does
Medical Patient Chart	`/residents` (modal)	Doctor, Admin	Longitudinal clinical encounter history, physician notes, ICD-10 impressions, prescriptions
Maternal Care (ANC)	`/maternal`	Nurse, Doctor, Admin	Prenatal enrollment with obstetric history (GPAL), 4-visit ANC tracking, high-risk flagging
Child EPI	`/immunizations`	BHW, Nurse, Doctor, Admin	DOH Expanded Program on Immunization schedule tracking and defaulter alerts
Hospital Referrals	`/reports/referral`	Nurse, Doctor, Admin	Generate standardized RF-1 referral slips for transfer to MHO or Abra Provincial Hospital

Governance & Administration

Page	Route	Roles	What It Does
Epidemiology Analytics	`/analytics`	Doctor, LGU Viewer, Admin	FHSIS morbidity surveillance dashboard with barangay health profile reports
Operations & Backup	`/operations`	LGU Viewer, Admin	Workstation diagnostics, localStorage cache status, Supabase cloud sync
User Management	`/users`	Admin only	Create accounts, assign roles, configure per-user module permissions (RBAC)

Nurse Workflow — Taking Vitals & Triage

This is the core clinical workflow. The nurse (or BHW) captures vital signs and the system automatically evaluates them for clinical flags.

Step 1: Select the Patient

Navigate to `/vitals` (Hemodynamic Screener & Triage). Search for the patient by name or UPI from the resident registry.

Step 2: Enter Vital Signs

The nurse enters the following measurements. Values outside the accepted range are rejected as physiologically implausible.

Vital Sign	Unit	Accepted Range
Systolic Blood Pressure	mmHg	40 – 300
Diastolic Blood Pressure	mmHg	30 – 200
Heart Rate	bpm	30 – 250
Respiratory Rate	cpm	8 – 80
Temperature	°C	30.0 – 45.0

Vital Sign	Unit	Accepted Range
Weight	kg	0.5 – 300
Height	cm	30 – 250

Step 3: Automatic Clinical Evaluation

The moment the nurse submits, the system runs `evaluateVitals()` which performs four checks in sequence:

1. **Blood pressure classification** — classifies using the Philippine Clinical Practice Guidelines for Hypertension into Normal, Pre-Hypertension, Stage 2 Hypertension, or Hypertensive Crisis
2. **Temperature screening** — checks if temperature ≥ 38.0 °C (febrile threshold)
3. **BMI calculation** — computes weight ÷ height² (in meters) and classifies using WHO Asia-Pacific cut-offs
4. **Overall triage severity** — assigns the highest severity found: NORMAL → ELEVATED → HIGH → CRITICAL

The system also sets clinical flags (e.g., `HYPERTENSIVE_CRISIS_EMERGENCY`, `FEBRILE_ILLNESS_ALERT`, `BMI_OBESE_2`) and recommends mandatory referral if the patient is in hypertensive crisis.

Step 4: Review and Save

The system displays a color-coded severity badge, all clinical flags, BMI classification, and whether a referral is recommended. The nurse can add clinical notes before saving the encounter.

Clinical Thresholds & Illness Detection

These are the exact thresholds programmed into the system. When a vital sign crosses a threshold, the system raises a clinical flag and may escalate the triage severity.

Blood Pressure Classification

Based on the Philippine Clinical Practice Guidelines for Hypertension. The system checks systolic first, then diastolic. The higher severity wins — for example, if systolic is 130 (pre-hypertension) but diastolic is 95 (stage 2), the patient is classified as Stage 2.

Classification	Systolic (mmHg)	Diastolic (mmHg)	Severity	Clinical Flag	Referral?
Normal	< 120	AND < 80	NORMAL	—	No
Pre-Hypertension	120 – 139	OR 80 – 89	ELEVATED	`PRE_HYPERTENSION`	No
Stage 2 Hypertension	140 – 179	OR 90 – 119	HIGH	`STAGE_2_HYPERTENSION`	No
Hypertensive Crisis	⩾ 180	OR ⩾ 120	CRITICAL	`HYPERTENSIVE_CRISIS_EMERGENCY`	YES (mandatory)

Temperature / Fever Screening

The febrile threshold is 38.0 °C. If a patient is febrile and their blood pressure severity is NORMAL, the overall triage severity is elevated to ELEVATED.

Reading	Classification	Flag
< 38.0 °C	Afebrile (normal)	—
⩾ 38.0 °C	Febrile	`FEBRILE_ILLNESS_ALERT`

BMI Classification (WHO Asia-Pacific)

The system uses WHO Asia-Pacific cut-offs, which are lower than Western standards. A BMI of 23 is already "overweight" and 25 is "obese" — compared to 25 and 30 in Western guidelines. BMI is auto-calculated as weight (kg) ÷ height² (m).

BMI Range	Classification	Flag
< 18.5	Underweight	`BMI_UNDERWEIGHT`
18.5 – 22.9	Normal	—
23.0 – 24.9	Overweight	— (monitored, no flag)
25.0 – 29.9	Obese Class I	`BMI_OBESE_1`
⩾ 30.0	Obese Class II	`BMI_OBESE_2`

Heart Rate & Respiratory Rate

These vitals are recorded and stored but the system does **not** automatically classify them as normal or abnormal. They are validated only for physiological plausibility (HR: 30–250 bpm, RR: 8–80 cpm). Clinical interpretation is left to the attending physician or nurse. There is no SpO2 field in the system.

What Triggers an Automatic Referral

Only one condition triggers a mandatory referral recommendation: **Hypertensive Crisis** (systolic ⩾ 180 OR diastolic ⩾ 120). All other flags (fever, BMI abnormality, pre-hypertension) are informational alerts for the clinician's judgment.

Maternal Care & High-Risk Flags

Enrollment

A nurse enrolls a pregnant resident into the Maternal Care module at `/maternal`. She enters the obstetric history using the GPAL system:

- **G (Gravida)** — total number of pregnancies including the current one
- **P (Para)** — number of deliveries past 20 weeks of gestation
- **A (Abortion)** — number of pregnancy losses before 20 weeks
- **L (Living Children)** — number of currently living children
- **Full Term / Preterm** — breakdown of delivery outcomes
- **LMP (Last Menstrual Period)** — used to calculate the estimated delivery date

Estimated Date of Delivery (EDD)

Calculated automatically using Naegele's Rule: LMP + 1 year − 3 months + 7 days. For example, if LMP is January 15, 2026, the EDD is October 22, 2026.

Gestational Age

Calculated as completed weeks from LMP to the current date.

High-Risk Pregnancy Flags

The system automatically flags these conditions at enrollment:

Condition	Threshold	Flag
Adolescent Pregnancy	Mother's age < 18 years	`ADOLESCENT_PREGNANCY`
Advanced Maternal Age	Mother's age ≥ 35 years	`ADVANCED_MATERNAL_AGE`
Grand Multiparity	Gravida ≥ 5 OR Para ≥ 5	`GRAND_MULTIPARITY`

ANC (Antenatal Care) 4-Visit Tracking

The system tracks 4 prenatal visits per pregnancy. Each visit records:

- Visit number (1 through 4)
- Gestational age in weeks
- Fundic height (cm)
- Fetal heart tone (bpm)
- Blood pressure (systolic and diastolic)
- Weight (kg)
- Iron-folic acid distribution (yes/no)
- Clinical notes

The analytics dashboard tracks the 4-visit ANC completion rate as a key maternal health indicator.

Child Immunization (EPI) & Defaulter Tracking

DOH EPI Schedule

The system follows the official Department of Health Expanded Program on Immunization. 14 vaccine doses are tracked across the first year of life:

Age	Vaccines Due
At birth	BCG (tuberculosis), Hepatitis B (within 24 hours)
6 weeks	Pentavalent 1 (DTP-HepB-Hib), Oral Polio 1, Pneumococcal 1
10 weeks	Pentavalent 2, Oral Polio 2, Pneumococcal 2

Age	Vaccines Due
14 weeks	Pentavalent 3, Oral Polio 3, IPV (Inactivated Polio), Pneumococcal 3
9 months	Measles-Rubella (MR)
12 months	MMR Booster

Defaulter Detection

The system compares each child's age in weeks against the EPI schedule. If a child is older than the target age and the vaccine has not been administered, they are flagged as a defaulter. The alert includes:

- Which vaccine is overdue and its name
- How many weeks overdue
- The child's barangay and purok (for BHW community outreach)

This enables targeted follow-up by BHWs who can visit the household directly.

Fully Immunized Child (FIC)

A child is considered "fully immunized" when they have received all 9 mandatory vaccines: BCG, Hepatitis B, Pentavalent 1–3, OPV 1–3, and Measles-Rubella. The FIC rate is reported in the barangay health profile analytics.

Morbidity Surveillance & Analytics

The Epidemiology Analytics page (`/analytics`) generates a Barangay Health Profile by scanning all vital sign records and classifying them into disease categories.

Disease Detection Heuristics

Condition	How the System Detects It
Hypertensive Crisis	Severity is CRITICAL, or SBP ≥ 180, or DBP ≥ 120
Stage 2 Hypertension	SBP ≥ 140 or DBP ≥ 90 (and not crisis level)
Febrile Illness	Temperature ≥ 38.0 °C

Condition	How the System Detects It
Suspected Dengue	Febrile + clinical notes contain "dengue" or "petechiae"
Acute Respiratory Infection	Febrile + not classified as dengue (catch-all for other febrile cases)

What the Profile Includes

- **Population data** — total resident count per barangay with purok breakdown
- **Morbidity counts** — number of cases per condition above
- **Maternal metrics** — total active pregnancies, 4-visit ANC completion count and coverage rate
- **Immunization metrics** — Fully Immunized Child (FIC) count and rate, defaulter line-list

This data supports the FHSIS (Field Health Services Information System) reporting required by the DOH and enables evidence-based municipal health planning.

Referral Slip Generation

The Hospital Referrals page (`/reports/referral`) generates a standardized RF-1 referral slip for transferring patients to the Municipal Health Office or Abra Provincial Hospital.

Auto-Generated Referral Reasons

Condition	Generated Reason
CRITICAL severity	"EMERGENCY: Hypertensive Crisis — SBP ≥ 180 or DBP ≥ 120"
Pregnant patient	Maternal consultation referral
Other	Clinician enters the reason manually

The slip includes patient demographics, vital signs, clinical flags, and the referring health worker's details. It can be printed or exported for the receiving facility.

Admin Operations

User Management (`/users`)

- Create new user accounts with Employee ID and password
- Assign one of the 5 roles (BHW, Nurse, Doctor, LGU Viewer, Admin)
- Configure per-user module access — override the role's default modules to grant or restrict specific pages
- All changes are logged in the audit trail for RA 10173 (Data Privacy Act) compliance

Workstation Diagnostics (`/operations`)

- View localStorage cache size and record counts for each data type
- Trigger cloud sync to Supabase
- Run system diagnostics (connectivity, latency, cache integrity)
- Export data as JSON backup

Security & Audit

- Every data access and modification is logged in the `audit_logs` table with actor, action type, table, record ID, and metadata
- Row-Level Security (RLS) on all database tables ensures users can only access data their role permits
- Module access is enforced client-side by the `ModuleGuard` component and server-side by Supabase RLS policies
- The admin always retains access to User Management regardless of module configuration